ELECTRIAL AND ELECTRONIC ENGINEERING

GROUP ONE

1. SMART BIN WITH GARBAGE LEVEL MONITORING

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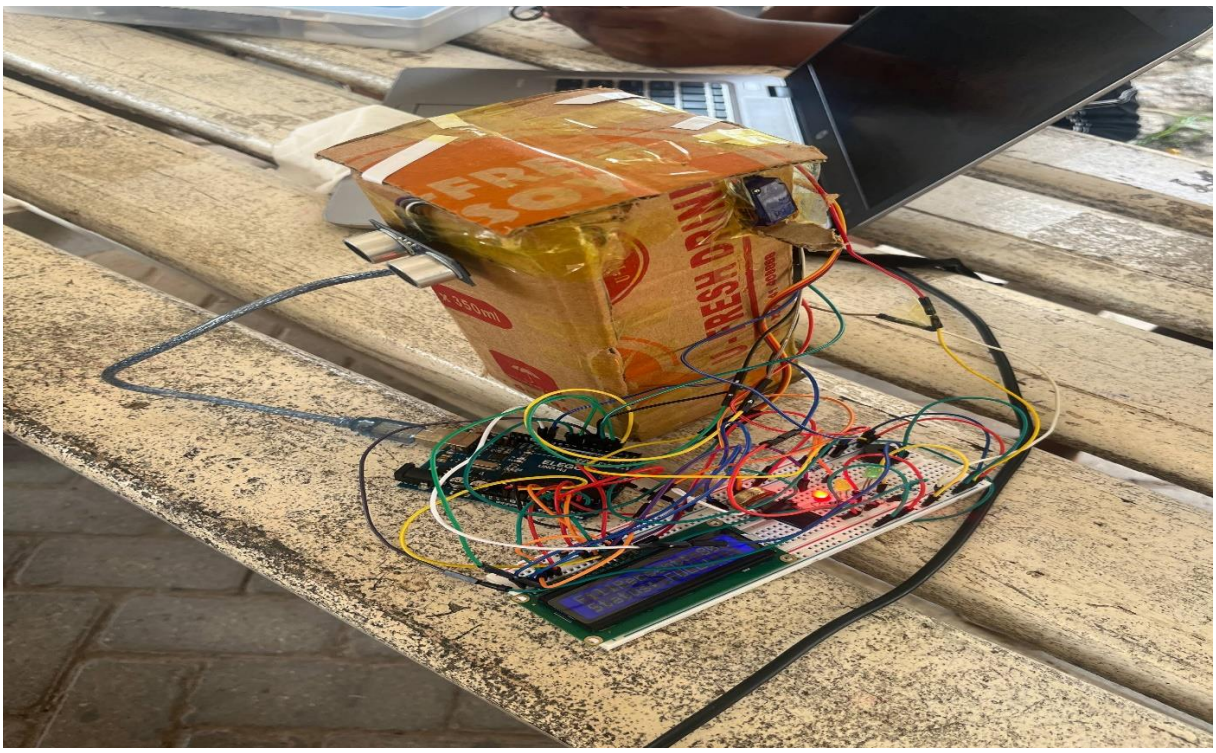

Figure 1: Smart Bin Prototype

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3.Problem statement

3.1 The Problem

Overflowing dustbins are a common issue in daily life, creating multiple challenges:

- **Unhygienic conditions:** Spread of germs, unpleasant odors, and health risks.
- **Wasted time:** Workers may check bins unnecessarily, lowering efficiency.
- **Environmental pollution:** Scattered waste contributes to litter and contamination.
- **Inefficient collection:** No reliable system to optimize waste collection schedules.
- **Lack of real-time monitoring:** Authorities and users cannot track bin status instantly.

In short, conventional bins are inefficient, unhygienic, and provide no data for better waste management.

Figure 2:Poorly maintained public bin area

3.2 The Solution

The **Smart Dustbin** addresses these problems with an intelligent, automated system:

- **Automatic lid opening:** Sensors detect approaching users and open the lid hands-free.
- **Real-time garbage monitoring:** Ultrasonic sensors track the fill level continuously.
- **LCD display:** Shows current fill percentage for quick assessment.
- **Alerts:** Colored LEDs and a buzzer notify users when the bin is full.

By combining automation, monitoring, and alerts, the Smart Dustbin ensures hygiene, efficiency, and convenience.

3.3 Project Objectives

Objective	Status
Automatically detect a person and open the lid	Achieved
Continuously measure garbage level	Achieved
Display fill percentage on LCD	Achieved
Indicate bin status with LEDs	Achieved
Trigger buzzer alert when full	Achieved

All project objectives have been successfully implemented, demonstrating a fully functional Smart Dustbin system.

4. Methodology

The methodology explains the **approach used to solve the problem of overflowing bins** and provides a **detailed overview of the Smart Bin's schematic and code logic**.

4.1 Approach to Solving the Problem

The Smart Bin project was designed to **automate waste management** and provide **real-time monitoring** of bin fill levels. The approach involves:

1. **Human Detection:**
    - An ultrasonic sensor detects when a person approaches the bin.
    - This triggers the servo motor to open the lid automatically, ensuring **hands-free operation**.
2. **Garbage Level Monitoring:**
    - The ultrasonic sensor continuously measures the distance from the lid to the top of the garbage pile.
    - This allows calculation of the bin's **fill percentage** in real time.
3. **Status Indication:**
    - A 16x2 LCD displays the fill percentage for easy monitoring.
    - Green, yellow, and red LEDs indicate empty, near-full, and full states, respectively.
    - A buzzer alerts users when the bin is full, preventing overflow.
4. **Automation and Feedback Loop:**
    - All sensors and actuators are integrated via the Arduino Uno.
    - The system continuously reads sensor data, updates displays and LEDs, and operates the servo motor automatically.

This approach ensures hygiene, efficiency, and smart waste management by addressing the problem of overflowing bins with minimal human intervention.

4.2 Circuit Schematic Overview

The schematic of the Smart Bin system shows the **connections between components**:

- **Arduino Uno** – central controller managing sensors, LEDs, servo, LCD, and buzzer.
- **Ultrasonic Sensor (HC-SR04)** – connected to Arduino digital pins for distance measurements.
- **Servo Motor (SG90)** – connected to a PWM pin for automated lid movement.
- **LCD 16x2 Display** – connected with data pins and a potentiometer for contrast adjustment.
- **LEDs (Green, Yellow, Red) and Buzzer** – connected via 330Ω resistors to Arduino digital pins.
- **Power Supply** – 5V from Arduino to power all components, with common GND.

The schematic ensures that all components work together seamlessly to automate bin monitoring and alerts.

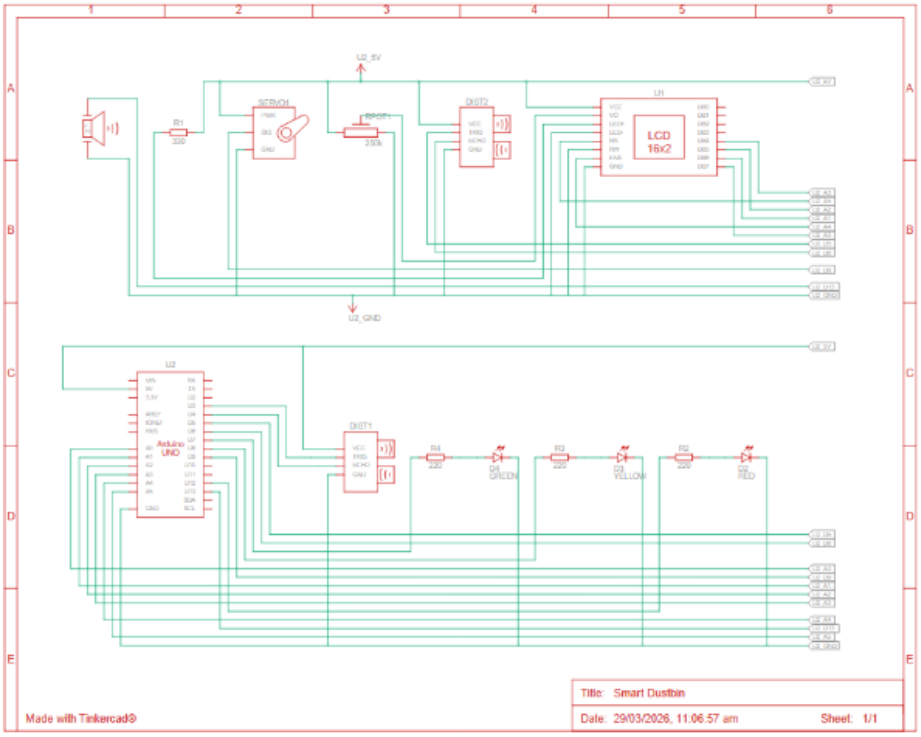

Figure 3:Smart Bin Circuit Schematic

4.3 Code Overview

The Arduino code controls all operations of the Smart Bin. The structure is as follows:

1. **Setup Section:**
    - Initializes pins for sensors, LEDs, buzzer, servo, and LCD.
    - Positions the servo motor at the default closed lid position.
2. **Main Loop:**
    - Continuously measures distance from the ultrasonic sensor to detect approaching persons.
    - Opens the servo-controlled lid when a person is detected within a set distance.
    - Monitors garbage level and calculates fill percentage.
    - Updates LCD with the current fill level.
    - Activates LEDs based on thresholds:
        - Green: empty or safe
        - Yellow: nearly full
        - Red: full
    - Sounds buzzer when the bin reaches full capacity.
3. **Key Functions:**

Function	Purpose
measureDistance()	Reads ultrasonic sensor and calculates distance to detect people and bin level
updateLCD()	Displays fill percentage on LCD in real time
controlLEDsAndBuzzer()	Changes LED colors and triggers buzzer based on bin status
operateServo()	Opens/closes the lid automatically when a person is detected

The code ensures continuous monitoring and automatic operation, integrating sensor inputs with actuator outputs.

4.4 Summary of Methodology

- **Problem Solved:** Overflowing bins, manual monitoring, and unhygienic disposal.
- **Solution Implemented:** Automated lid operation, real-time monitoring, LED and buzzer alerts.
- **Integration:** Arduino Uno coordinates sensors, actuators, LCD, and alert mechanisms.
- **Outcome:** Fully functional Smart Bin that operates automatically and provides instant feedback to users

5. Demonstration / Simulation

The demonstration showcases the functionality of the Smart Bin using **simulation and real-time observation**.

Steps in the Demonstration

1. **Simulation Setup:**
    - The Smart Bin circuit is set up on a breadboard or in an Tinkercad.
    - All components — Arduino Uno, ultrasonic sensor, servo motor, LCD, LEDs, and buzzer — are connected as per the schematic.
2. **Person Detection Test:**
    - When a person approaches the bin, the ultrasonic sensor detects the distance.
    - The **servo motor opens the lid automatically**, demonstrating **hands-free operation**.
3. **Bin Fill Monitoring:**
    - The ultrasonic sensor continuously measures the garbage level.
    - The **LCD displays the real-time fill percentage**.
4. **Alerts and Indicators:**
    - **Green LED** lights when the bin is empty.
    - **Yellow LED** lights when the bin is near full.
    - **Red LED** lights and the **buzzer sounds** when the bin is full.

5. **Observation of Functionality:**
    - Each component responds correctly to sensor inputs.
    - The system demonstrates **automation, monitoring, and user alerts** successfully.

Figure 4: Smart Bin simulation in Tinkercard

6.Key Takeaways

From the Smart Bin demonstration and simulation, the following key points are observed:

- **Hands-Free Operation:** The bin lid opens automatically when a person approaches, ensuring hygiene.
- **Real-Time Monitoring:** The fill percentage is displayed on the LCD, allowing users to know the bin status instantly.
- **Effective Alerts:** LEDs and buzzer provide clear visual and audible signals when the bin is near full or full.
- **Automation Works Seamlessly:** The Arduino integrates all components, demonstrating efficient coordination between sensors, actuators, and indicators.
- **Problem Solved:** Overflowing bins are prevented, and manual monitoring is no longer required.
- **Scalability:** The system can be expanded to multiple bins or connected to IoT for real-time monitoring across locations.

7.Conclusion

The Smart Bin project successfully demonstrates an **automated waste management system** that addresses the problem of overflowing bins while promoting hygiene and efficiency.

7.1 Findings

- The bin **automatically opens its lid** when a person approaches, providing hands-free operation.
- Real-time **garbage level monitoring** is displayed on the LCD, allowing users to check fill percentage instantly.
- **LED indicators and buzzer alerts** effectively signal the bin's status (empty, near-full, full).
- The system operates **reliably and consistently**, integrating sensors, actuators, and displays with the Arduino Uno.

7.2 Challenges Encountered

- **Sensor calibration:** Adjusting the ultrasonic sensor to accurately detect both people and garbage levels.
- **Servo motor control:** Ensuring smooth lid movement without jerking or overshooting.
- **Circuit organization:** Managing multiple components on a breadboard to prevent loose connections.

7.3 Possible Improvements

- **IoT Integration:** Connect bins to a network to monitor fill levels remotely.
- **Multiple Bin Management:** Expand the system to monitor several bins simultaneously.
- **Solar Power:** Make the system energy-efficient for outdoor deployment.
- **Enhanced Alerts:** Add mobile notifications for maintenance staff when bins are full.

Overall, the Smart Bin system provides a practical, efficient, and hygienic solution to the problem of overflowing bins, demonstrating the potential of sensor-based automation in waste management

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9. Appendix

Figure 5:Group 01-Smart Bin project Team